BUI TRAN TAN PHAT

AI ENGINEERING INTERN
GENAI & VISION-LANGUAGE MODELS

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EDUCATION

HCMC University of Technology and Engineering - HCMUTE

- B.E. in Information Technology, Specialization in Artificial Intelligence — Expected August 2027
- GPA: **3.5/4.0**

RESEARCH & PROJECT EXPERIENCE

Skin-Lesion Classification on HAM10000

- Engineered and integrated ~30 handcrafted visual features with clinical patient metadata to construct comprehensive training matrices.
- Trained and fine-tuned LightGBM classifiers via randomized search, achieving an Accuracy of 0.807 and an OvR ROC-AUC of 0.946 on severely imbalanced data.

High-Resolution Retinal Vessel Segmentation — Research Project

- Developed hybrid UNet++ model with EfficientNet-B4 backbone to segment retinal vessels, targeting thin micro-vessels that traditional methods often miss.
- Applied data augmentation and evaluated with Dice, IoU, Sensitivity, and Accuracy; achieved Dice 0.771 and Accuracy 0.964 on HRF dataset.

EmoViS: Fine-Grained Vietnamese Emotion Recognition — Co-author, accepted at AMI 2026

- Co-developed dual-branch model combining ViSoBERT with 300-dimensional Emoji2Vec embeddings for 28-label emotion recognition on ViGoEmotions (20,664 Vietnamese social media comments).
- Ran ablation over 15+ configurations; reached 0.6329 Macro-F1 with three-seed ensemble, a significant gain of +0.0156 over reproduced baseline (0.6173; paired bootstrap, $p < 0.0001$).
- Isolated emoji branch as only individually significant component (+0.0198 Macro-F1, $p < 0.0001$); asymmetric loss and class-balanced reweighting yielded no significant gain.

PUBLICATIONS

- N.-P. Nguyen, T.-T.-P. Bui, G.-H. Huynh, V.-Q. Hoang, “EmoViS: Emoji-Augmented ViSoBERT for Fine-Grained Multi-Label Emotion Recognition in Vietnamese Social Media Text,” accepted at the International Conference on Autonomous Machine Intelligence (AMI) 2026.

TECHNICAL SKILLS

- Programming Languages: Python.
- ML/DL Frameworks: PyTorch.
- Databases: MySQL, ChromaDB.
- Data & Computer Vision: Pandas, NumPy, OpenCV, Matplotlib, Seaborn.

SKILLS

- Programming: Python
- ML/DL: PyTorch, LightGBM
- Computer Vision & Data: OpenCV, Pandas, NumPy, Matplotlib, Seaborn
- Databases: MySQL, ChromaDB
- Languages: Vietnamese (native), English (IELTS 6.0)